PENGTENG LI

Ph.D. Candidate · Embodied AI

Vision-Language-Action Models · Spatial Perception, Understanding & Modeling

The Hong Kong University of Science and Technology (Guangzhou)

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ACHIEVEMENTS

- **Industry & Open-Source —** Co-founded **AlphaBrain Platform** (GitHub, 180+ ⭐, MIT), AI² Robotics' open-source embodied-AI framework; **lead developer of the world-model-based VLA stack** (V-JEPA 2.1 / Cosmos / Wan). Featured by **Synced (机器之心)**, **QbitAI (量子位)** & 10+ top AI media — **100k+ reads**.
- **Research — Embodied AI First —** Flagship first-author **ICML 2026** (SOMA: +30% real-robot success, SOTA on RoboCasa & SimplerEnv); co-author on the first brain-inspired VLA, **under review at a Nature family journal**; cross-venue output across CoRL, ICRA, IROS, IJCAI, NeurIPS, ACM MM, TMM. **6 first-author CCF-A papers**; *~400 citations, h-index 7.*

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou), Ph.D. in Computer Science — *Sep 2024 – Jul 2027 (expected)*

Thrust of Artificial Intelligence · Advisor: Prof. Hui Xiong (AAAS / IEEE / ACM Fellow)

Shenzhen University, M.Sc. in Computer Science — *Sep 2021 – Jun 2024*

GPA 3.63 / 4.0 (Top 1%) · Advisor: Prof. F. Richard Yu (FRSC / MAE / IEEE Fellow) · Outstanding Graduate; National Scholarship

INDUSTRY & RESEARCH EXPERIENCE

AI² Robotics (智平方) · *¥10B+ valuation*, Research Intern — *May 2025 – Present*

Co-Lead, Core VLA Research (Shenzhen) · Co-Founder, AlphaBrain Platform

VLA & world-model-based policy learning · Real-world bimanual manipulation · Teleoperation data pipeline

Supervised by Yandong Guo

- Flagship research: led SOMA (ICML 2026, first author) — +30% real-world manipulation success over GR00T-N1.5 / Diffusion Policy on out-of-vision tasks; co-authored NeuroVLA — the first brain-inspired VLA (third author, under review at a Nature family journal), owning experimental design and paper layout for the team's spiking-VLA work.
- Co-founded **AlphaBrain Platform** (GitHub repo, 180+ stars, MIT) — AI² Robotics' open-source embodied-intelligence framework. Owned the core system architecture and the world-model-based VLA stack (V-JEPA 2.1 / Cosmos 2 / 2.5 / Wan 2.2). **Industry impact:** covered by 10+ top AI / tech media — incl. **Synced (机器之心)** and **QbitAI (量子位)** — **100k+ reads**.
- End-to-end data-collection quality lead: owned quality control across the full demo-collection pipeline on Bot1s / Bot2 — covering teleop validation, data-evaluation standards and reproduction workflow — substantially uplifting downstream VLA training quality and real-world manipulation performance.

Tencent Youtu Lab, Computer Vision Algorithm Intern — *Jan 2024 – Apr 2024*

- Research: adapted open-vocabulary object detection methods to the few-shot object-detection regime, improving rare-class transfer on internal benchmarks.
- Production: validated PCB-defect-detection requirements for an enterprise client using PatchCore and related anomaly-detection algorithms, supporting the front-end product team's deployment pipeline.

Guangming Laboratory, Algorithm Engineer (Research Intern) — *Aug 2022 – Dec 2023*

- Foxconn project — designed and validated an automated iPhone self-inspection production line and built mobile-board key-component recognition on top of in-house smart glasses.
- Shenzhen SDG project — built a few-shot pipeline for road-defect detection and recognition under complex urban environments.
- DeepRoute project — advanced 3D-perception frameworks for autonomous-driving scenarios, including occupancy prediction and multi-view fusion.

FEATURED RESEARCH

SOMA: Spatial Memory for Out-of-Vision Manipulation in Vision-Language-Action *ICML 2026 · Seoul*

Pengteng Li, *Weiyu Guo, He Zhang, Tiefu Cai, Xiao He, Yandong Guo*, Hui Xiong* · HKUST(GZ) × AI² Robotics · first author*

- Identified a fundamental view-bound limitation of existing VLAs (GR00T-N1.5, SpatialVLA, OpenVLA, π0): policies fail when task-relevant objects fall outside the camera frustum.
- Designed a memory-centric VLA framework — Spatial Memory Construction (YOLO + DINOv3 + VGGT multi-view fusion) + Dynamic Memory Refinement (similarity-aware EMA update) + Contextual Memory Retrieval (cross-attention into the DiT action decoder).
- Real-world: +30% average SR over GR00T-N1.5 on 5 OOV tasks (single-arm, multi-step, dual-arm), with 40–60% drops in fixation time, head-search length and grasp attempts — i.e. near one-shot grasping.
- Simulation: 52.0% SR on RoboCasa GR1 tabletop (300 demos, +7.7 pts vs. GR00T-N1.5); 63.2% on SimplerEnv Visual Matching — new SOTA.

See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Models *NeurIPS 2025 · San Diego*

Pengteng Li, *Pinhao Song, Wuyang Li, Huizai Yao, Weiyu Guo, Yijie Xu, Dugang Liu, Hui Xiong · HKUST(GZ) · first author*

- First training-free framework that enhances the spatial-understanding ability of multimodal LLMs (GPT-4V, Qwen-VL, LLaVA family) by combining off-the-shelf perception models with maximum-semantic-richness sampling and motion reconstruction.
- Plug-and-play and resource-efficient — no fine-tuning required; runs CPU-only and slots into any MLLM as a modular prompting layer.
- Mitigates the well-known spatial-reasoning bottleneck of vision-only inputs at the prompt level; consistent gains across spatial QA, scene understanding and grounding benchmarks, paving the perception groundwork for downstream VLA models such as SOMA.

SELECTED PUBLICATIONS

*# first author · * corresponding author · underline = VLA / embodied-AI works*

Robotics & Embodied AI (VLA, Manipulation, Navigation)

- **[ICML 2026]** **SOMA: Spatial Memory for Out-of-Vision Manipulation in Vision-Language-Action.** **#**
  P. Li, *W. Guo, H. Zhang, T. Cai, X. He, Y. Guo*, H. Xiong*.*
- **[Nature Family · Under Review]** **A Brain-inspired Embodied Intelligence for Fluid and Fast Reflexive Robotics Control.**
  W. Guo, H. Zhang, ***P. Li****, T. Cai, Z. Chen, Y. Guo, X. He, Y. Yang, Y. Sun et al. · first brain-inspired VLA*
- **[CoRL 2024]** **Implicit Grasp Diffusion: Bridging Dense Prediction and Sampling-Based Grasping.**
  P. Song, ***P. Li****, R. Detry. · 9 citations*
- **[ICRA 2024]** **Robot Trajectron: Trajectory-Prediction-Based Shared Control for Robot Manipulation.**
  P. Song, ***P. Li****, E. Aertbeliën, R. Detry. · 23 citations*
- **[IROS 2024]** **A Language-Driven Navigation Strategy Integrating Semantic Maps and LLMs.**
  Z. Zhong, Y. He, ***P. Li****, F. Yu, F. Ma. · 5 citations*

- **[arXiv 2025]** [Equivariant Volumetric Grasping.]
  *P. Song, Y. Hu, **P. Li**, R. Detry. · 1 citations*
- **[IJCAI 2024]** [ABM: Attention before Manipulation.]
  Collaboration. · 4 citations
- **[IROS 2024]** [LLaKey: Follow My Basic Action Instructions to Your Next Key State.]
  Collaboration. · 3 citations

Perception & Spatial Understanding

- **[NeurIPS 2025]** [See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Models.] **#**
  ***P. Li**, P. Song, W. Li, H. Yao, W. Guo, Y. Xu, D. Liu, H. Xiong. · 7 citations*
- **[IJCAI 2024]** [OTOcc: Optimal Transport for 3D Occupancy Prediction.] **#**
  ***P. Li**, et al.*
- **[AAAI 2026]** [Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection.]
  *H. Yao, S. Zhao, **P. Li**, Y. Cui, S. Lu, W. Guo, Y. Lu, Y. Xu, H. Xiong. · 1 citations*
- **[ACM MM 2023]** [IGG: Improved Graph Generation for Domain Adaptive Object Detection.] **#**
  ***P. Li**, Y. He, F. R. Yu, P. Song, D. Yin, G. Zhou. · 12 citations*
- **[ICASSP 2023]** [Bagging R-CNN: Ensemble for Object Detection in Complex Traffic Scenes.] **#**
  ***P. Li**, Y. He, D. Yin, F. R. Yu, P. Song. · 7 citations*
- **[Neurocomputing 2023]** [Boosting R-CNN: Reweighting R-CNN Samples by RPN's Error for Underwater Object Detection.]
  *P. Song, **P. Li**, L. Dai, T. Wang, Z. Chen. · 307 citations*
- **[arXiv 2025]** [You Only Need 4 Extra Tokens: Synergistic Test-Time Adaptation for LLMs.]
  Collaboration. · 2 citations
- **[AAAI 2025]** [Subgraph Invariant Learning Towards Large-scale Graph Node Classification.]
  Collaboration. · 2 citations

Event Cameras (MLLM Understanding & 3D Reconstruction)

- **[TMM 2026 · Major Revision]** [EventVL: Understand Event Streams via Multimodal Large Language Models.] **#**
  ***P. Li**, Y. Lu, P. Song, W. Li, H. Yao, H. Xiong. · 12 citations*
- **[TMM 2026]** [DeblurSplat: SfM-Free 3D Gaussian Splatting with Event Camera for Robust Deblurring.] **#**
  ***P. Li**, Y. Lu, P. Song, W. Guo, H. Yao, F. R. Yu, H. Xiong.*
- **[arXiv 2025]** [From Events to Enhancement: A Survey on Event-Based Imaging Technologies.]
  *Y. Lu, X. Xu, **P. Li**, Y. Wang, Y. Cui, H. Yao, H. Xiong.*
- **[IJCV 2026]** [SEE: See Everything Every Time — Adaptive Brightness Adjustment for Broad Light Range Images via Events.]
  Collaboration. · 2 citations
- **[CVPR 2026]** [From Events to Clarity: The Event-Guided Diffusion Framework for Dehazing.]
  Collaboration. · 1 citations

Other

- **[NeurIPS 2025]** [Multimodal 3D Genome Pre-training.]
  *M. Yang, **P. Li**, Y. Liang, Q. Cai, Z. Zheng, S. Zhang, P. Zhang, Z.-A. Huang, et al.*

HONORS & SELECTED ACTIVITIES

- Outstanding Graduate, Shenzhen University, 2024.
- National Scholarship (top graduate student award), China Ministry of Education, 2023.
- Reviewer for top-tier venues in computer vision, robotics and machine learning.

TECHNICAL SKILLS

Embodied AI Stack: VLA frameworks (GR00T-N1.5, OpenVLA / OpenVLA-OFT, π0 / π0-FAST, SpatialVLA, StarVLA, Diffusion Policy); world models (V-JEPA 2.1, Cosmos 2 / 2.5, Wan 2.2); benchmarks (RoboCasa / 365, SimplerEnv, LIBERO / LIBERO-plus); real-robot stack (Realman ZM73 dual-arm, RealSense D435, Jetson AGX Orin, Quest 3 + GELLO teleop, ROS 2).

ML & Multimodal: PyTorch, JAX, HuggingFace Transformers / Diffusers; DiT, flow matching, RLHF / DPO; vision-language models (CLIP, LLaVA, Qwen-VL); 3D Gaussian Splatting, 3D occupancy, VGGT, DINOv3, YOLO, event-camera perception; distributed training on 32+ H100 / H200 with FlashAttention, DeepSpeed / FSDP.

Engineering & Languages: Python (primary), C++, CUDA basics, Linux, Docker, Git, AWS / cluster scheduling. Mandarin (native), English (professional working — international publications and conference presentations).